

क्रम सं/SL No :044191121



सत्यमेव जयते

पेटेंट कार्यालय, भारत सरकार

The Patent Office, Government Of India

पेटेंट प्रमाण पत्र

Patent Certificate

(पेटेंट नियमावली का नियम 74)

(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.

546396

आवेदन सं. / Application No.

202141029702

फाइल करने की तारीख / Date of Filing

01/07/2021

पेटेंटी / Patentee

INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IIT Madras)

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित **AUTOMATED COVERSLIPPER FOR LARGE FORMAT SLIDES WITH SWITCHABLE COMPATIBILITY TO HANDLE MULTI FORMAT SLIDES** नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख जुलाई 2021 के पहले दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled **AUTOMATED COVERSLIPPER FOR LARGE FORMAT SLIDES WITH SWITCHABLE COMPATIBILITY TO HANDLE MULTI FORMAT SLIDES** as disclosed in the above mentioned application for the term of 20 years from the 1st day of July 2021 in accordance with the provisions of the Patents Act, 1970.



अनुदान की तारीख : 30/07/2024
Date of Grant :

पेटेंट नियंत्रक
Controller of Patents

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, जुलाई 2023 के पहले दिन को और उसके पश्चात प्रत्येक वर्ष में उसी दिन देय होगी।
Note - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 1st day of July 2023 and on the same day in every year thereafter.

[Home \(https://ipindia.gov.in/\)](https://ipindia.gov.in/)
[About Us \(https://ipindia.gov.in/Home/AboutUs\)](https://ipindia.gov.in/Home/AboutUs)
[Policy & Programs \(https://ipindia.gov.in/Home/policypages\)](https://ipindia.gov.in/Home/policypages)
[Achievements \(https://ipindia.gov.in/Home/achievementspage\)](https://ipindia.gov.in/Home/achievementspage)
[RTI \(https://ipindia.gov.in/Home/righttoinformation\)](https://ipindia.gov.in/Home/righttoinformation)
[Sitemap \(https://ipindia.gov.in/Home/Sitemap\)](https://ipindia.gov.in/Home/Sitemap)
[Contact Us \(https://ipindia.gov.in/Home/contactus\)](https://ipindia.gov.in/Home/contactus)

[Skip to Main Content](#)

[\(http://ipindia.nic.in/index.htm\)](http://ipindia.nic.in/index.htm)

 [\(http://ipindia.nic](http://ipindia.nic)

Patent Search

Invention Title	AUTOMATED COVERSLIPPER FOR LARGE FORMAT SLIDES WITH SWITCHABLE COMPATIBILITY TO HANDLE MULTI FORMAT SLIDES
Publication Number	05/2023
Publication Date	03/02/2023
Publication Type	INA
Application Number	202141029702
Application Filing Date	01/07/2021
Priority Number	
Priority Country	
Priority Date	
Field Of Invention	PHYSICS
Classification (IPC)	G01N0001310000, G02B0021340000, A43B0003100000, A63G0021180000, G01N0001300000

Inventor

Name	Address	Country
Jayaraj Joseph	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India
Mohanasankar Sivaprakasam	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India
Jayaraman Kiruthi Vasan	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India
Ramdayalan K	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India
Rahul Manoj	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India
Sudhan Chandrasekaran	Indian Institute of Technology Madras, IIT Madras, Sardar Patel Road, Chennai 600 036, Tamil Nadu, India	India

Applicant

Name	Address	Country
INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IIT Madras)	The Dean, Industrial Consultancy & Sponsored Research [IC&SR], Indian Institute of Technology Madras, Sardar Patel Road, IIT P.O, Chennai, Tamil Nadu 600 036, India	India

Abstract:

A system for performing coverslipping includes multiple zones, where a coverslipper device is designated to move from one or more zones. An input zone comprises which comprises a stack of stained slides and a slide that is designated to be picked up by a slide pick-up platform. The picked-up slide reaches a dispensation zone a mountant medium is dispensed via a nozzle and the slide pickup platform moves to a coverslipping zone. The coverslipper device picks the cover slips from a slip picl and moves to a coverslipping zone. The slide pick-up platform positions itself in the coverslipping zone while the picked coverslip is mounted over the slide. The proci slipped slide is transported by a slide conveyor and inserted into a slide tray, which is moved up and down and is positioned to accommodate the next processed slid

Complete Specification

DESC:Attached ,CLAIMS:Attached

[View Application Status](#)



Terms & conditions (<https://ipindia.gov.in/Home/Termsconditions>) Privacy Policy (<https://ipindia.gov.in/Home/Privacypolicy>)
Copyright (<https://ipindia.gov.in/Home/copyright>) Hyperlinking Policy (<https://ipindia.gov.in/Home/hyperlinkingpolicy>)
Accessibility (<https://ipindia.gov.in/Home/accessibility>) Contact Us (<https://ipindia.gov.in/Home/contactus>) Help (<https://ipindia.gov.in/Home/help>)
Content Owned, updated and maintained by Intellectual Property India, All Rights Reserved.

Page last updated on: 26/06/2019



Industrial Consultancy & Sponsored Research (IC&SR)

AUTOMATED COVERSLIPPER FOR LARGE FORMAT SLIDES WITH SWITCHABLE COMPATIBILITY TO HANDLE MULTI FORMAT SLIDES

IITM Technology Available for Licensing

PROBLEM STATEMENT

- **Coverslipping** is a process where protective glass sheets are applied to glass slides with tissue specimens.
- The goal is to **protect the specimen** from external impacts and improve visibility by straightening the tissue.
- Current automated coverslippers can only handle smaller slide sizes, typically **25 x 75 mm (1" X 3")** slides.
- There's a need for an efficient way to automate coverslipping of large format histology slides (**150 X 200mm**) while maintaining compatibility with medium and small slides.
- Current commercial products can only process **1 X 3 slides**.
- A device controlled by a **custom-defined protocol logic** is needed for error-free coverslipping of large format slides.
- The device should be **capable of handling stacks of large format slides** and continuously perform the process.
- An improved coverslipping and dispensing technique is needed to ensure **uniform distribution of mountant medium** and minimize air gaps.

TECHNOLOGY CATEGORY MARKET

Technology: Automated coverslipper to handle multi format slides

Category: Biotechnology & Genetic Engineering

Industry: Biomedical

Application: Tissue engineering

Market: The global market size was valued at **USD 184.5 Million in 2023** and is projected to reach **USD 259.2 Million by 2030**, growing at a **CAGR of 5.1%** during the forecast period **2024-2030**.

INTELLECTUAL PROPERTY

IITM IDF Ref. 2202 ,Patent No: IN 546396
PCT/IN2022/050598

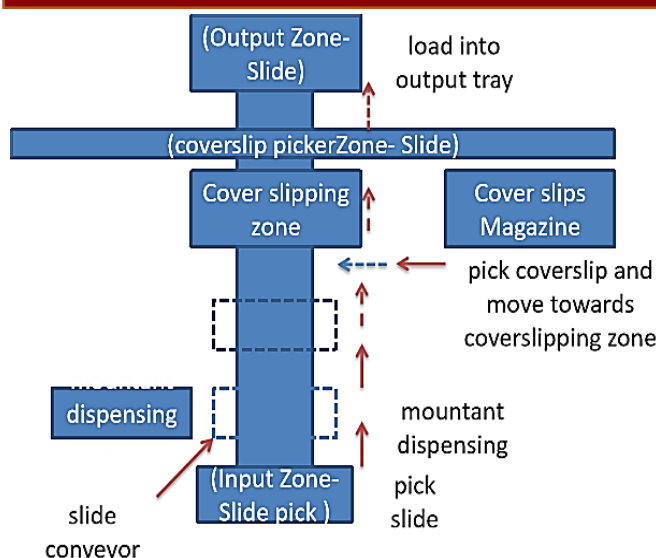
TRL (Technology Readiness Level)

TRL- 4, Experimentally validated in Lab;

Research Lab

Prof. Jayaraj Joseph & Prof. Mohanasankar sivaprakasam,
Dept. of Electrical Engineering

TECHNOLOGY



1 • A coverslipping system consists of a coverslipper device on a frame that moves from one or more zones.

2 • The input zone has a slide tray with stained slides, which the coverslipper device picks up from a platform.

3 • The dispensation zone dispenses a mountant medium, and the slip pick-up zone picks up the coverslips from the tray.

4 • The coverslipping zone positions the platform in the coverslipping zone, and the picked coverslip is mounted over the slide.

5 • The output zone transports the processed cover slip slide and inserts it into a tray, which is moved up and down to accommodate the next processed slide.

CONTACT US

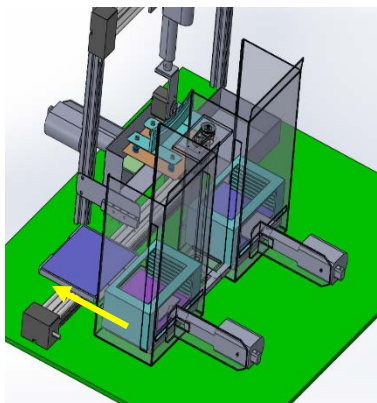
Dr. Dara Ajay, Head TTO
Technology Transfer Office,
IPM Cell- IC&SR, IIT Madras

IITM TTO Website:
<https://ipm.icsr.in/ipm/>

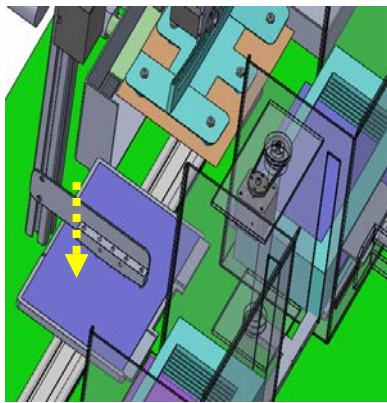
Email: headtto-icsr@icsrpis.iitm.ac.in
tto-mktg@icsrpis.iitm.ac.in

Phone: +91-44-2257 9756/ 9719

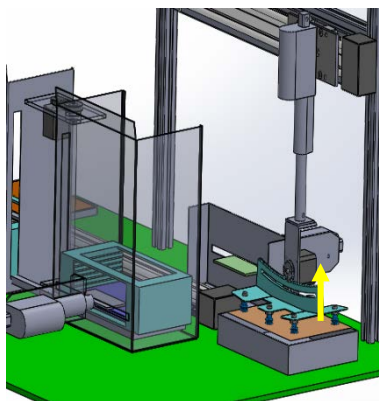
Fig 1 shows the sequence of operations using the slide transport system that is associated with the coverslipper



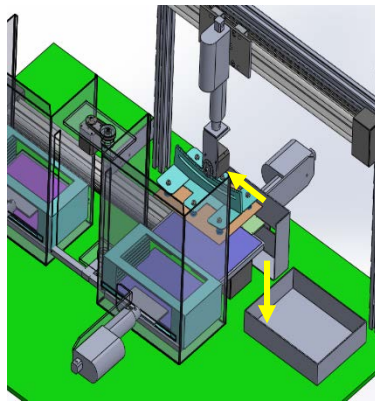
(1 a) Loading of glass slide on conveyor



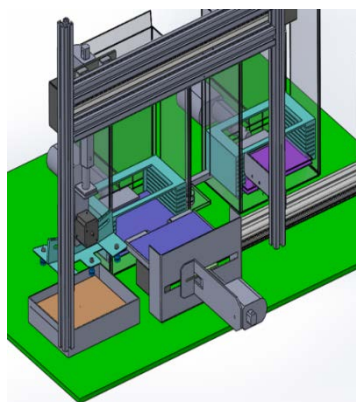
(1 b) Dispensing mountant on slide



(1 c) Picking up cover slip



(1 d) Positioning & cover slipping



(1 e) Unloading the slide into output tray

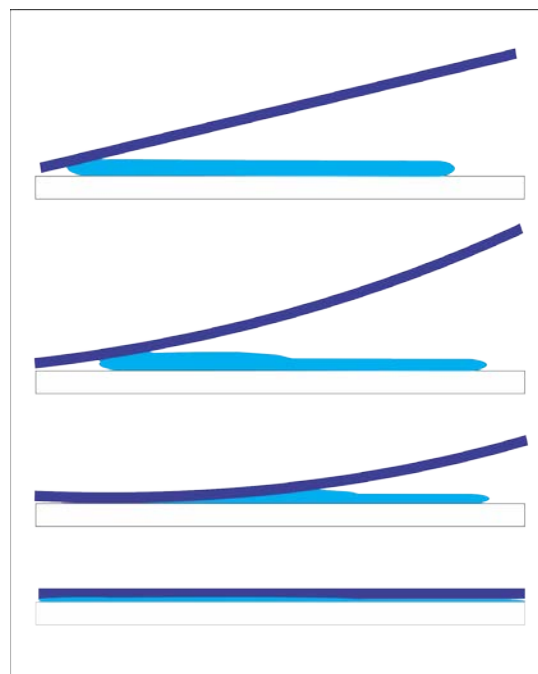


Fig 2 Cover-slip technique

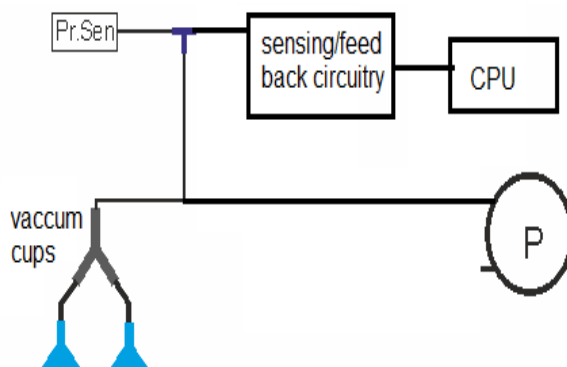


Fig 4: Pressure sensing circuitry

Fig 3: 60mm wedge nozzle

CONTACT US

Dr. Dara Ajay, Head TTO
Technology Transfer Office,
IPM Cell- IC&SR, IIT Madras

IITM TTO Website:
<https://ipm.icsr.in/ipm/>

Email: headtto-icsr@icsrpis.iitm.ac.in
tto-mktg@icsrpis.iitm.ac.in

Phone: +91-44-2257 9756/ 9719



IIT MADRAS

Indian Institute of Technology Madras

Technology Transfer Office
TTO - IPM Cell



Industrial Consultancy & Sponsored Research (IC&SR)

Fig 5 illustrates an electromechanical system control unit associated with the system architecture of the coverslipper device

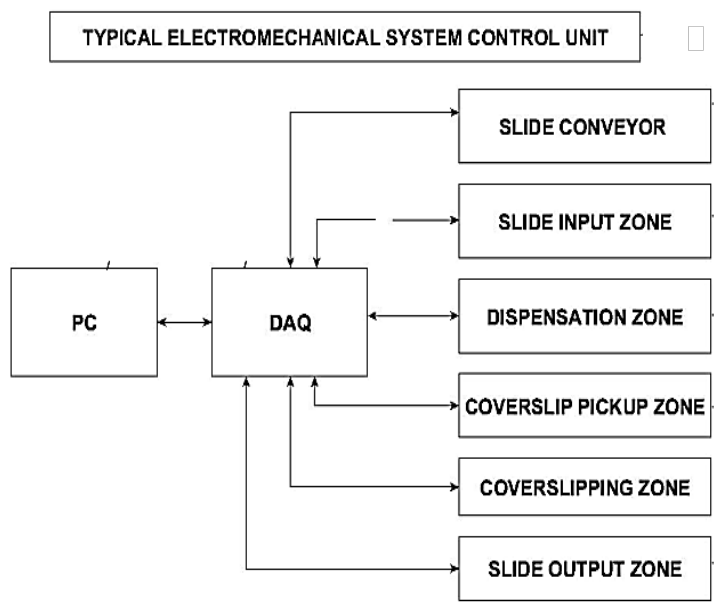


Fig 6:
1 stripe
of each 8
to 9ml of
mounting
medium



Fig 7:
Output
after
cover-
slipping
- no air
bubbles

Dispersing axis
Cover-slipping axis(start to end)

Key Features / Value Proposition

Coverslipper Device Overview

- Processes large format slides of 6" X 8" and 5" X 7", and can be customized to smaller sizes.
- Features a slide tray with stained slides inserted into grooves for easy pick-up. Uses a wedge-shaped nozzle to uniformly dispense mountant medium over the slide.
- Controlled by a dedicated algorithm for accurate and uniform placement and air bubble prevention.
- Composed of a vertical plate, a first curved plate, and a second curved plate.
- Rotates the second curved plate along a predefined path, allowing vacuum suction cups to lift the cover slip.
- Achieves a unique locus of motion to prevent air pockets and bubbles below the cover slip. The rack mimics human ankle motion between Dorsiflexion and Plantarflexion positions.

Uses and Applications:

- Prevents contact between the microscope's objective lens and the specimen.
- Provides an even thickness (in wet mounts) for viewing depth.
- Viewing enhancement as the specimen is flattened.
- Deceleration of evaporation from the sample, both in wet and dry mounted slides.
- Permanent affixation for long term and repeated use of Permanent Specimens.

Large Format Specimen Protection and Storage

- Enables long-term preservation and storage of large format specimens.
- Economizes cover-slipping process through indigenous algorithms
- Versatile, supporting multiple slide sizes, a unique feature not found anywhere else
- Supports cover-slipping of large quantities of slides, like adult human brain sections.
- Processes large format cover-slips, a unique feature not developed globally.

CONTACT US

Dr. Dara Ajay, Head TTO
Technology Transfer Office,
IPM Cell- IC&SR, IIT Madras

IITM TTO Website:
<https://ipm.icsr.in/ipm/>

Email: headtto-icsr@icsrpis.iitm.ac.in
tto-mktg@icsrpis.iitm.ac.in

Phone: +91-44-2257 9756/ 9719